

becomes void. What do you do in a situation like this?

Absolutely Nothing!

You take the news like an adult, remain calm and stoic, and move on. You still have that horrible feeling—the kind that feels like someone just ripped out your intestines—but hey, when technology goes wrong, it does so horribly. Life goes on.

What you *can* do, however, is learn from the mistakes of others, and try and safeguard yourself from ever having to be in this situation. My mind is a lot more at ease now because I've taken a few steps to prevent such disasters. If you have data that's really important to you, you should seriously consider following in my footsteps.

RAID

You've heard it a million times, everyone has heard about RAID, but how many of us have found out enough to actually use it?

RAID, or Redundant Array of Inexpensive Disks, can either be used to improve performance or to back up your data. The most common forms are RAID 0 (Striping) and 1 (Mirroring). Striping improves performance, because it uses two disks as one. Basically, it breaks up data into two and writes one part to one disk and the other to the second disk. This results in faster performance, which is great because the hard drive is the bottleneck in all current-generation PCs. Of course, this also doubles your risk of losing data, because even if one disk dies, you lose all the data on both disks. Only gamers and those who need faster hard drives usually use RAID 0.

RAID 1, or Mirroring, may even result in a performance hit, but creates redundancy. Say you have two 250 GB hard drives, and set them in Mirroring mode. Windows will see them as one 250 GB hard drive. RAID 1 works by writing all data twice—once each to the two drives. What this does is create two copies of everything, so say if one hard drive dies, your system will just continue running like nothing ever happened. Beautiful, isn't it?

Most of us don't use RAID because it requires you to have two hard drives, preferably identical. So you end up spending double the money, and end up with a quite ridiculous cost-per-GB figure. However, after the three recent incidences of hard drives playing foul that I came across, an extra Rs 4,000 to mirror my existing 250 GB drive is worth it for the peace of mind I now have.

If you decide to use RAID 1, I suggest you keep your operating system on a third drive, so that formatting and reinstalling or virus attacks cannot threaten to wipe out your data. For example, I have an 80 GB drive that holds my OS and download folders, while my dual-250 GBs are in RAID 1, and hold everything that's dear to me. The advantage is that I find the performance of my PC to be a little better, since the OS just reads and writes to my 80 GB, and mainly reads from the mirrored drives (when I'm playing audio or video or viewing pictures), and sometimes writes to both (when I back up important stuff). This helps to ensure that both

disks last me a long time.

Stuff to remember: Check if your motherboard has RAID controllers—most motherboards made in the last two years will. Next, go out and buy two identical drives—although not necessary, it will save you from the potential headaches of mismatched sizes.

Check your motherboard manual for instructions on how to set up RAID, or if you're buying a separate RAID

controller, follow the instructions given in the manual that accompanies it. Remember, for motherboards with onboard RAID controllers, you will need to enable it from the BIOS.

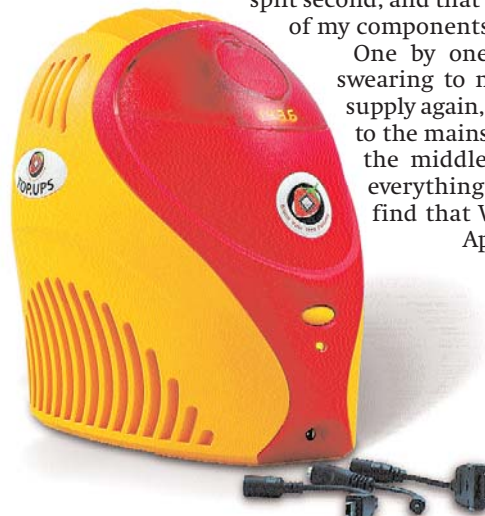
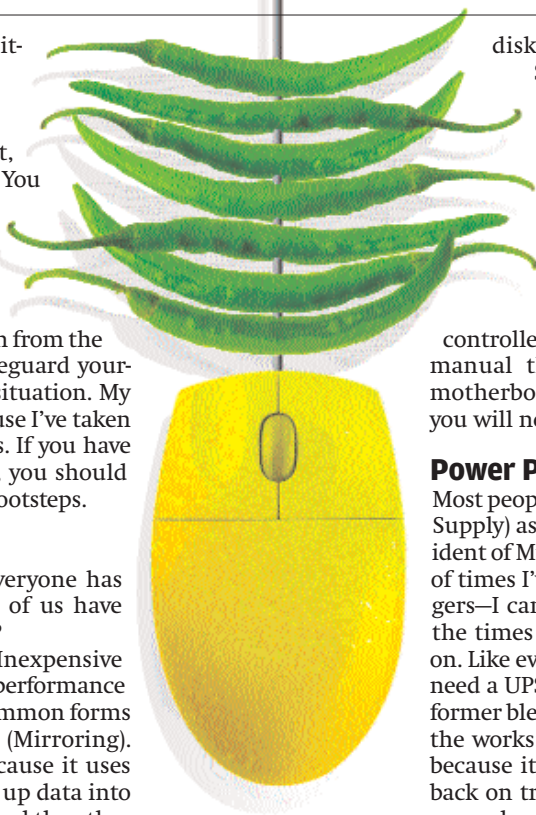
Power Problems

Most people think of a UPS (Uninterrupted Power Supply) as a backup source of power. Being a resident of Mumbai, I can literally count the number of times I've witnessed power outages on my fingers—I can actually use just one hand to count the times the power's gone out and my PC was on. Like everyone else, I used to think I just don't need a UPS. About two years ago, the local transformer blew up—literally, with sparks, flames and the works; quite exciting that, and really pretty because it happened at night... Anyway, getting back on track, I was working on a write-up, had some downloads running, and had my regular applications running—anti-virus, image editor, MS Word, MS Excel, a media player, a few IM applications, etc. When everything went black, I was actually relieved, because I figured I'd take a long overdue break from writing, and went out to see what all the commotion was about.

A few hours later when the transformer was repaired / bypassed / <whatever they did to fix it>, the power supply resumed. The first thing I noticed was a dark patch around the back of my SMPS. Pressing the power button on my PC confirmed my fears—a blown power supply. A day or two later, I went out and bought a new SMPS, only to come back and find that the motherboard also seemed to be fried. Apparently, just before the transformer blew up, it sent out way over the normal 240 volts for a split second, and that seemed to have fried a lot of my components.

One by one I replaced fried parts, swearing to never buy a cheap power supply again, or connect my PC directly to the mains without a spike board in the middle. Three days later, I got everything working again, only to find that Windows refused to load.

Apparently the power surge had scrambled data on my hard drive as well, because although it was being detected in the BIOS, it refused to admit to having Windows XP. A re-install later, I realised I'd lost all my downloads, my work, and all the settings and programs I'd arranged the way I like it.



UPS' aren't just boring old boxes anymore. See the Bazaar section for more on this product!